Robust Allocentric Spatial Reasoning in LLMs under Underspecified Map Instructions

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Abstract

Natural-language spatial instructions are often inherently underspecified, requiring inference from partial or implicit information. However, large language models (LLMs) are typically evaluated in fully specified settings that do not reflect this ambiguity.

Using the Rendezvous (RVS) dataset (Paz-Argaman et al., 2024), we generate 21,049 controlled variants of human-written allocentric instructions by masking landmarks and directional terms while keeping the underlying map fixed. A symbolic oracle assigns *Answerable*, *Ambiguous*, or *Contradictory* labels based on constraint satisfiability. Across models, FLAN-T5 collapses toward predicting *Answerable* and OLMo toward *Ambiguous*, with both showing near-zero detection of *Contradictory* cases and performing below a majority-class baseline.

Despite these failures in answerability classification, models still produce geographically plausible goal predictions (Success@250m: 39–41%), indicating that spatial grounding and constraint tracking can diverge. Overall, the results suggest that current LLMs show limited adaptation to missing spatial cues and tend to default to model-specific output priors rather than adjusting their predictions to the underlying constraint structure.

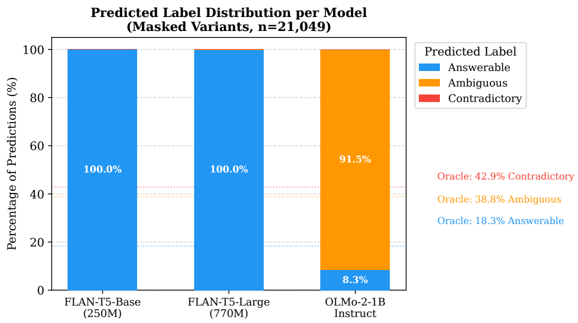

Figure 1: Predicted label distributions across all three models on masked variants ($n = 21{,}049$). FLAN-T5 collapses to *Answerable* while OLMo collapses to *Ambiguous*. Neither model produces meaningful *Contradictory* predictions. This collapse occurs regardless of how much spatial information is removed.

1 Introduction

Spatial reasoning underlies everyday communication about locations and their relationships, yet such descriptions are often *underspecified*: speakers omit landmarks, rely on implicit directional cues, and assume shared map knowledge. Allocentric spatial reasoning (inferring locations from relations among landmarks rather than from egocentric step-by-step navigation) is especially sensitive to such omissions. It plays a central role in navigation, robotics, and instruction-following systems (Yang et al., 2025; Kong et al., 2025), where agents must correctly resolve spatial relations to act on natural-language instructions.

Prior work has shown that large language models struggle with spatial reasoning even under fully specified conditions (Shi et al., 2022; Li et al., 2024; Paz-Argaman et al., 2024). However, existing evaluations typically assess accuracy under fixed inputs and do not examine how model predictions change when critical spatial information is selectively removed. This gap is particularly important for allocentric reasoning, where removing a single landmark or directional cue can transform an instruction from solvable to ambiguous or contradictory.

To study this systematically, we use the Rendezvous (RVS) dataset (Paz-Argaman et al., 2024), which pairs human-written allocentric instructions with real urban map graphs and human-validated target locations. We generate controlled underspecified variants by masking landmarks, directions, or both while keeping the underlying map and goal fixed. A two-stage symbolic–neural oracle assigns *Answerable*, *Ambiguous*, or *Contradictory* labels based on constraint satisfiability. As shown in Figure 1, the evaluated models collapse to stable output distributions regardless of instruction content, suggesting limited sensitivity to the underlying constraint structure.

1.1 Research Question

We investigate the following question: **How robust are LLMs at allocentric spatial reasoning when natural-language descriptions of space are systematically underspecified, even though a consistent underlying world (urban map graph) exists?**

By isolating the effect of linguistic underspecification while holding the environment fixed, we aim to determine whether models adapt their predictions to missing spatial cues or default to model-specific output priors regardless of constraint structure.

2 Related Work

Allocentric spatial reasoning has emerged as a key testbed for evaluating whether language models can integrate linguistic descriptions with structured geographic knowledge (Wang et al., 2025; Qiao et al., 2025; Zhang et al., 2025). Prior work shows that while LLMs can often interpret individual spatial relations, they struggle to maintain globally consistent representations across multiple constraints.

Paz-Argaman et al. (2024) introduce the RVS dataset, a geodetically grounded benchmark pairing human-written allocentric instructions with real OpenStreetMap city graphs. Their results demonstrate that LLMs face difficulty with multi-scale spatial inference even when instructions are fully specified. Our work extends this setting by holding the environment fixed while systematically removing linguistic cues, enabling direct measurement of robustness to underspecification.

Robustness under compositional stress has been explored in synthetic environments. Shi et al. (2022) propose StepGame, where reasoning depth is varied to stress-test multi-hop spatial inference. Performance degrades as compositional complexity increases, highlighting limitations in constraint composition. In contrast, we vary the *completeness* of spatial descriptions rather than their depth, focusing on information removal in realistic map-based settings.

More broadly, Li et al. (2024) show that LLMs often recognize individual spatial relations but fail to compose them into coherent global structures. However, these evaluations primarily assess accuracy under fixed input conditions and do not examine how predictions change when critical information is removed, the gap our framework directly addresses.

Together, these works establish that spatial reasoning in LLMs is brittle under both compositional and informational stress. Our contribution is a controlled evaluation framework that quantifies this brittleness in a geospatially grounded setting using a symbolic oracle independent of model behavior.

3 Methodology

We evaluate allocentric spatial reasoning under controlled underspecification using a hybrid symbolic–neural oracle. Natural-language instructions are mapped to symbolic spatial constraints, grounded to map entities, and resolved against an urban graph. The design separates constraint satisfiability (symbolic) from semantic matching (neural), ensuring that oracle labels reflect linguistic structure rather than model stochasticity.

3.1 Symbolic Resolution Pipeline

The oracle converts each instruction into a constraint triple (category, landmark name, direction) using deterministic extraction rules. This projection onto a single landmark and direction is a deliberate scoping choice: it isolates the effect of removing one constraint type at a time, making each masking operation directly interpretable. Multi-relation instructions are reduced to their primary constraint; modeling compound constraint interactions is left to future work.

Semantic grounding. Extracted phrases are matched against pre-encoded POI embeddings using `all-MiniLM-L6-v2` (Reimers and Gurevych, 2019): $\text{sim}(q, p) = \langle f_\theta(q),\ f_\theta(p)\rangle$. Neural grounding is invoked only when symbolic exact matching fails, extending coverage to paraphrastic expressions (e.g., "place where you work out" $\rightarrow$ GYM) without allowing embeddings to override spatial constraint satisfaction. A small category prior ($+0.10$) rewards exact OSM tag matches, following the dense retrieval approach of Karpukhin et al. (2020) adapted to structured map graphs.

Spatial filtering. Grounded candidates are filtered using radius-based proximity (1,500m, following RVS findings on Manhattan's spatial density), eight-directional wedge filtering consistent

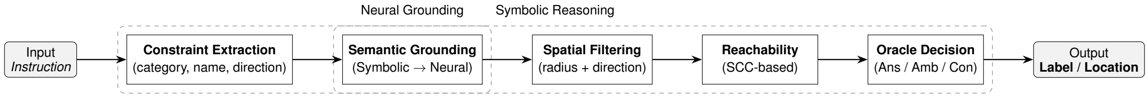

Figure 2: Hybrid symbolic–neural oracle pipeline. Constraint extraction, spatial filtering, and the oracle decision are fully deterministic. Semantic grounding uses `all-MiniLM-L6-v2` as a fallback when symbolic matching fails, ensuring oracle labels reflect constraint structure rather than embedding similarity. The decision procedure assigns *Answerable*, *Ambiguous*, or *Contradictory* based solely on the cardinality of surviving graph candidates.

with human cardinality bias (Paz-Argaman et al., 2024), and SCC-based reachability pruning to remove topologically disconnected nodes.

Decision procedure. Let C be the set of surviving candidates. The oracle assigns: *Answerable* if $|C| = 1$, *Ambiguous* if $|C| > 1$, and *Contradictory* if $|C| = 0$. All steps are deterministic; the neural fallback does not affect the final candidate count.

3.2 Two-Stage Oracle Framework

Oracle 1 (filtering mode). Applied to original RVS instructions before masking. When multiple candidates survive, Oracle 1 applies salience-based disambiguation using the RVS LANDMARK hierarchy: `wikipedia` > `wikidata` > `brand` > `tourism` > `amenity` > `shop`, returning binary labels (*Answerable* or *Contradictory*). This produces a symbolically verified, humanly-navigable subset from which controlled degradation can be measured.

Oracle 2 (labeling mode). Applied to masked variants and original instructions without salience collapse, returning three-way labels based solely on $|C|$.

Design choice. An alternative design would apply Oracle 2 directly to original instructions. This would label most originals as *Ambiguous*, technically correct under strict uniqueness but misleading: original RVS instructions carry no *Ambiguous* label by construction, as they were human-validated as navigable. Solver ambiguity on originals likely reflects symbolic imprecision rather than genuine linguistic underspecification. We therefore use Oracle 1 to construct the filtered subset, and Oracle 2 exclusively for three-way evaluation of masked variants and the originals derived from that subset. The tradeoff is that results are specific to the symbolically resolvable subset of RVS.

3.3 Data Construction and Dataset Statistics

The data construction pipeline proceeds as follows: original RVS instructions are filtered by Oracle 1, masked under three conditions (landmark, direction, both), and re-labeled by Oracle 2. Because the map graph and gold target remain fixed throughout, all variation in oracle labels is attributable to linguistic underspecification alone. Table 1 reports the resulting dataset statistics.

Split	Examples	Cities
Original (Oracle-1 filtered)	9,301	3
Manhattan	7,000	
Philadelphia	1,278	
Pittsburgh	1,023	
Masked variants (Oracle-2 labeled)	21,049	3
mask_landmark	7,163	
mask_directions	6,943	
mask_both	6,943	

Table 1: Dataset statistics after Oracle-1 filtering and variant generation. Each original instruction yields up to three masked variants; some variants are omitted when the corresponding element is absent in the instruction.

3.4 Oracle Consistency and Internal Validation

We do not have human re-annotation of oracle labels (see §7). However, several internal consistency checks support the oracle's reliability.

Cross-oracle stability. Table 2 shows the joint Oracle-1/Oracle-2 distribution on original instructions (n = 9,301). The *Contradictory* class is perfectly stable: all 2,138 Oracle-1-Contradictory examples remain *Contradictory* under Oracle-2 (100.0% agreement), confirming that both oracles share the same underlying constraint resolution. The 96.7% of Oracle-1-Answerable examples relabeled *Ambiguous* by Oracle-2 reflects the intended effect of removing salience disambiguation.

End-to-end sanity checks. A 50-sample subset run on OLMo confirmed non-degenerate oracle output across all three label classes prior to full-scale evaluation. POI category matching was verified on the Manhattan index (GARDEN: 744 matches, CAFE: 776 matches, 20,979 total embeddings). All evaluations use greedy decoding (`do_sample=False`) ensuring full reproducibility.

	Oracle-2 Label			
Oracle-1	*Ans.*	*Amb.*	*Con.*	**Total**
Answerable	235	6,928	0	7,163
Contradictory	0	0	2,138	2,138
Total	235	6,928	2,138	9,301

Table 2: Joint Oracle-1 $\times$ Oracle-2 distribution on original instructions ($n = 9{,}301$). Oracle-1 returns binary labels only; Oracle-2 returns three-way labels. The *Contradictory* class is perfectly stable (100.0%), and 96.7% of Oracle-1-Answerable examples are relabeled *Ambiguous* by Oracle-2.

4 Experimental Design

4.1 Tasks, Models, and Baselines

We consider two tasks. The **primary task** is answerability classification, where models predict *Answerable*, *Ambiguous*, or *Contradictory* for each instruction, evaluated against Oracle 2 labels. The **secondary task** is goal localization: predicted destinations are resolved to graph nodes and scored against the gold target using haversine distance (Success@250m, Success@100m).

We evaluate three instruction-tuned models: FLAN-T5-base (250M; Chung et al. 2022), FLAN-T5-large (770M), and OLMo-2-0425-1B-Instruct (1B; Groeneveld et al. 2024).

For answerability, we use a **majority-class baseline** (always *Contradictory*, 42.9%). For localization, we use a **nearest-POI baseline** selecting the closest POI of the oracle-extracted category to the start location. We further analyze in-context learning (ICL) effects on OLMo via zero-shot and 4-shot settings, including sensitivity to demonstration order.

4.2 Experimental Setup

All models use 4-shot in-context learning with greedy decoding (`do_sample=False`). Demonstrations are fixed across models, drawn from the Oracle-1 filtered subset and formatted as `<instruction, label>` pairs.

For OLMo, we additionally evaluate zero-shot, 4-shot (original order),and 4-shot (shuffled) conditions; unparseable zero-shot outputs are counted as incorrect. Experiments are conducted on both Oracle-2-labeled masked variants and their corresponding original instructions. Code is available at `https://github.com/Adan-Assi/nlp-allocentric-spatial-reasoning`.

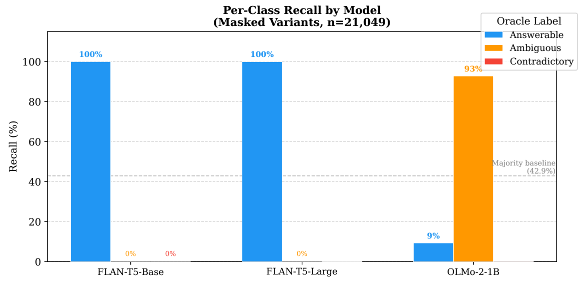

Figure 3: Per-class recall by model on masked variants ($n = 21{,}049$). FLAN-T5 variants achieve near-perfect *Answerable* recall (100%) but zero recall on both other classes. OLMo-2-1B shows a qualitatively different pattern: high *Ambiguous* recall (92.8%) driven by its *Ambiguous* prediction prior, and near-zero *Contradictory* recall (0.2%). No model achieves meaningful recall on *Contradictory* inputs under any condition.

5 Experimental Results

5.1 Answerability Classification

Table 3 and Figure 3 summarize model performance on masked variants ($n = 21{,}049$). All models fall below the majority-class baseline (42.9%). FLAN-T5-base and FLAN-T5-large collapse almost entirely to predicting *Answerable*, achieving $\approx$100% recall on that class and 0% on the others (macro-F1: 0.103). The two models are indistinguishable across all metrics despite a 3$\times$ parameter difference, suggesting that scale alone does not address this failure mode. OLMo-2-1B exhibits the opposite failure mode, predicting *Ambiguous* for nearly all inputs (92.8% recall, macro-F1: 0.228). Across all models, *Contradictory* recall is effectively zero. OLMo's *Contradictory* F1 of 0.004 reflects 55% precision on only 29 total predictions of that class; the model almost never assigns this label. Appendix Figure 10 provides full confusion matrices for all three models, and Appendix Figure 11 reports per-city accuracy patterns, confirming that these behaviors are consistent across geographic regions.

Error analysis. Table 4 shows OLMo's full confusion matrix. The dominant error is Oracle=*Contradictory* predicted as *Ambiguous* ($n = 8{,}195$, 38.9% of all predictions): when no valid candidate exists, OLMo almost always predicts multiple valid solutions. The second most common error is Oracle=*Answerable* predicted as *Ambiguous* ($n = 3{,}488$): OLMo hedges on uniquely solvable instructions. Together, these two error types account for 55.4% of all predictions and are consis-

Model	Acc.	Ans. R	Amb. R	Con. R	Mac. F1
Majority class	42.9	—	—	—	—
FLAN-T5-base	18.3	100.0	0.0	0.0	0.103
FLAN-T5-large	18.3	99.9	0.0	0.2	0.103
OLMo-2-1B	37.8	9.3	92.8	0.2	0.228

Table 3: Answerability classification on masked variants ($n = 21{,}049$). All values are percentages; majority-class baseline always predicts *Contradictory* (42.9%). FLAN-T5's *Answerable* F1 of 0.309 reflects 100% recall at baseline precision (18.3%), indicating total label collapse. OLMo's *Contradictory* F1 of 0.004 reflects 55% precision on only 29 predictions.

Oracle \ Pred.	*Ans.*	*Amb.*	*Con.*
Answerable	**359**	3,488	3
Ambiguous	576	**7,581**	10
Contradictory	819	8,195	**16**

Table 4: OLMo-2-1B confusion matrix on masked variants ($n = 21{,}047$ parseable). Diagonal entries (bold) are correct predictions. Oracle=*Contradictory* predicted as *Ambiguous* ($n = 8{,}195$) is the dominant error, accounting for 38.9% of all predictions and confirming that OLMo's *Ambiguous* prior overrides constraint structure.

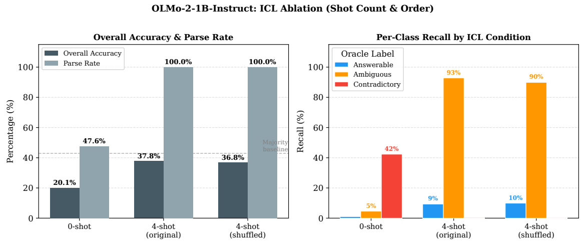

Figure 4: OLMo-2-1B-Instruct answerability accuracy and parse rate under three ICL conditions on masked variants. Zero-shot prompting produces a 47.6% parse failure rate and a qualitatively different recall pattern (Contradictory recall: 42.3% vs. 0.2% under 4-shot). Moving from zero-shot to 4-shot restores full parseability and raises accuracy by 17.7 percentage points. Example order has negligible effect (Δ1.0pp).

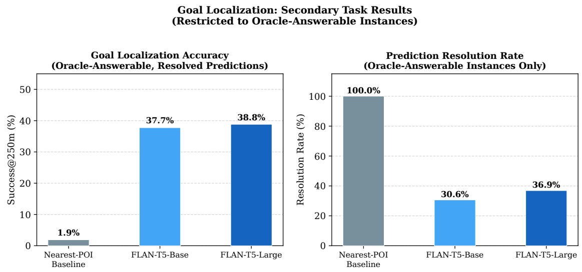

Figure 5: Goal localization results on masked variants. FLAN-T5 models substantially outperform the nearest-POI baseline on resolved predictions (S@250m: 39–41% vs. 2.8%; S@100m: 15–17% vs. 0.8%). Resolution rates of 34–38% show that a substantial fraction of outputs do not resolve to any graph node.

tent with a dominant *Ambiguous* bias that persists regardless of constraint structure.

Qualitative inspection confirms the pattern. An instruction such as "Meet me at the `[MASK]` on Elizabeth Street. A hostel is across the street to the `[DIR_MASK]` and an events venue is a block away" has oracle label *Contradictory* (no valid candidate survives masking), yet receives an *Ambiguous* prediction. An instruction with unique solution such as "I'm at a fast food restaurant on 7th Avenue, east of your location. This restaurant is on the `[MASK]` corner" receives *Ambiguous* despite being oracle-*Answerable*. In both cases OLMo's prediction is independent of the actual constraint structure. FLAN-T5's dominant error is Oracle=*Contradictory* predicted as *Answerable* ($n = 819$), the mirror-image collapse. Appendix Figure 8 visualizes this cross-condition stability directly.

5.2 In-Context Learning Ablation

Table 5 and Figure 4 report OLMo under three ICL conditions. Example order has negligible effect (Δ 1.0pp overall; Δ <3pp on any per-class recall). Shot count induces a qualitative shift: zero-shot prompting yields a 47.6% parse failure rate (11,040 of 21,049 outputs), dominated by fragments of the prompt's label definitions ("exactly one destination": 2,080 cases). Among parseable zero-shot outputs, accuracy is 42.3%, at the majority-class baseline, indicating no task understanding without demonstrations.

Critically, the per-class recall pattern under 0-shot differs qualitatively from 4-shot: *Contradictory* recall rises from 0.2% to 42.3% while *Ambiguous* recall collapses from 92.8% to 4.6%. OLMo predicts *Contradictory* only 29 times under 4-shot but 9,317 times under zero-shot, a shift in prediction prior, not instruction sensitivity. This inversion also reverses per-variant accuracy: `mask_directions` drops from 82.4% to 13.0% and `mask_both` rises from 11.7% to 33.9%, tracking the prior shift rather than the instruction content.

5.3 Goal Localization

Table 6 and Figure 5 report goal localization. FLAN-T5-base resolves 34.3% of outputs (7,218 of 21,049); FLAN-T5-large resolves 38.5% (8,104). On resolved predictions, Success@250m is 39.2% (base) and 41.2% (large), far exceeding the nearest-POI baseline (2.8% S@250m, 0.8% S@100m).

Condition	Acc.	Parse	Ans. R	Amb. R	Con. R	Pred Con.
4-shot (original)	37.8	100.0	9.3	92.8	0.2	29
4-shot (shuffled)	36.8	100.0	9.9	89.9	0.2	40
0-shot	20.1	47.6	1.0	4.6	42.3	9,317

Table 5: OLMo-2-1B-Instruct under three ICL conditions on masked variants (n = 21,049). Majority baseline: 42.9%. Parse rate is the fraction of outputs containing a valid label; 0-shot accuracy is over all examples. Per-class recalls use the full per-class support as denominator (Answerable: 3,850; Ambiguous: 8,169; Contradictory: 9,030) and need not sum to 100%.

Model	Res. Rate	S@250m	S@100m
Nearest-POI	100.0	2.8	0.8
FLAN-T5-base	34.3	39.2	15.7
FLAN-T5-large	38.5	41.2	17.1

Table 6: Goal localization on masked variants (n = 21,049). Resolution rate (%) is the fraction of model outputs mappable to a graph node; unresolved outputs are counted as failures. S@250m and S@100m are evaluated on resolved predictions only.

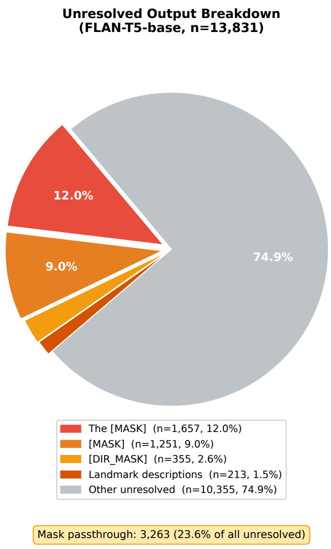

Figure 6: Breakdown of unresolved outputs for FLAN-T5-base (n = 13,831). Mask token passthrough (outputs where the model echoes `[MASK]`, `[DIR_MASK]`, or related fragments rather than generating a landmark name) accounts for 23.6% of all unresolved outputs (3,263 of 13,831). The dominant pattern is "The [MASK]" (1,657 cases, 12.0%), indicating the model treats the mask token as a valid referent rather than inferring the missing landmark.

Ambiguous examples resolve most frequently (40.7–43.6%) as they retain more intact landmark mentions; *Contradictory* examples achieve the highest localization accuracy (40.1% base, 43.5% large), confirming that geographic plausibility and constraint satisfiability are analytically independent.

Accuracy varies by category: HOSPITAL (53.5%), MARKET (52.4%), and BAR (50.4%) achieve the highest Success@250m, while BUILDING (18.3%) and WATER (20.0%) perform worst, consistent with geographic distinctiveness. Among unresolved outputs, mask token passthrough accounts for 23.6% (3,263 of 13,831 unresolved), with "The [MASK]" (1,657 cases) as the dominant pattern (Figure 6). Node-level category match rates are higher for geographically correct predictions than incorrect ones (base: 8.1% vs. 4.4%), confirming that Success@250m is a meaningful localization signal; full category analysis appears in Appendix Figure 7.

6 Discussion

6.1 Models Do Not Track Constraint Satisfiability

Across all evaluations, the tested models provide limited evidence of adjusting their predictions in response to changes in spatial constraint coverage. Per-class recalls remain nearly identical between masked and unmasked conditions despite large shifts in oracle label distributions. A model performing constraint-sensitive reasoning would exhibit at least a partial correlation between masking and predicted labels; instead, predictions remain largely unchanged. Appendix Figure 9 visualizes this invariance.

FLAN-T5's collapse to *Answerable* is consistent with its template-based instruction tuning, which encourages producing well-formed, resolvable outputs (Chung et al., 2022). OLMo's collapse to *Ambiguous* mirrors the dominant Oracle-2 label on original instructions (74.5%), suggesting RLHF post-training has internalized a distributional prior favoring hedged interpretations (Groeneveld et al., 2024).

6.2 Architecture and Post-Training Matter More Than Scale

FLAN-T5-base (250M) and FLAN-T5-large (770M) exhibit indistinguishable behavior across all metrics (accuracy: 18.3% each; macro-F1:

0.103 each), indicating that parameter count alone does not address the *Answerable* collapse. OLMo-2-1B (comparable in size to FLAN-T5-large) produces a qualitatively different failure mode and higher accuracy (37.8%, macro-F1: 0.228), suggesting that architecture (decoder-only vs. encoder-decoder) and post-training procedure (RLHF vs. template tuning) are more predictive of behavior than scale (Ouyang et al., 2022).

6.3 Localization and Answerability Are Separate Capabilities

FLAN-T5 achieves 39–41% Success@250m on resolved predictions despite failing answerability classification. Node-level category match rates are higher for geographically correct predictions than incorrect ones (base: 8.1% vs. 4.4%), confirming that geographic and semantic accuracy are positively correlated and that Success@250m is a meaningful localization signal. This pattern suggests that localization relies primarily on geographic memorization from pretraining, whereas answerability requires integrating relational constraints that the evaluated models do not appear to encode reliably. Evaluating both tasks is therefore necessary: reporting only localization accuracy would overstate spatial reasoning ability, while reporting only answerability failure would obscure genuine geographic competence.

7 Limitations

The symbolic oracle assumes a single target location, complete map coverage, and no pragmatic inference. Human interpreters rely on contextual salience and shared knowledge that our oracle does not model, potentially underestimating the solvability of some instructions. Our constraint extractor projects each instruction onto at most one landmark and direction, enabling controlled analysis but not capturing full relational complexity. This abstraction means that our conclusions reflect sensitivity to the oracle's simplified constraint structure rather than to the full set of relations expressed in natural allocentric instructions. We evaluate only three open-weight instruction-tuned models below 1B parameters; larger proprietary models may exhibit different behavior. The ICL ablation was conducted for OLMo only, as FLAN-T5's label collapse made it uninformative. Goal localization is evaluated only on predictions that resolve to a graph node (34–38% of outputs), introducing a selection effect independent of localization accuracy.

Finally, we do not evaluate oracle-human agreement. Some instructions labeled as *Ambiguous* or *Contradictory* by the oracle may be solvable by humans using pragmatic inference, meaning our results reflect oracle-defined solvability rather than human interpretation.

Our evaluation is limited to three U.S. cities from the RVS dataset, constraining the geographic diversity of tested environments. Broader geographic coverage may reveal different patterns of model behavior.

The node-level category match metric is constrained by OSM tag vocabulary: categories encoded under non-`amenity` keys (e.g., `shop`, `leisure`, `highway`) yield zero matches by construction, underestimating true category accuracy for those classes. A fuller evaluation would require a unified OSM-to-oracle category mapping, which we leave to future work.

8 Conclusion

We evaluated LLM robustness in allocentric spatial reasoning under systematically underspecified instructions. Using controlled masking over the RVS dataset and a deterministic symbolic oracle, we observed that the evaluated instruction-tuned models provide limited evidence of adjusting predictions in response to changes in spatial constraint coverage. FLAN-T5 collapses to *Answerable*, OLMo collapses to *Ambiguous*, and neither model reliably detects *Contradictory* cases. Despite this, models generate geographically plausible landmark names, revealing a separation between localization ability and constraint-sensitive reasoning. Our framework and results provide a controlled benchmark for future work on spatial reasoning under underspecification.

A Appendix

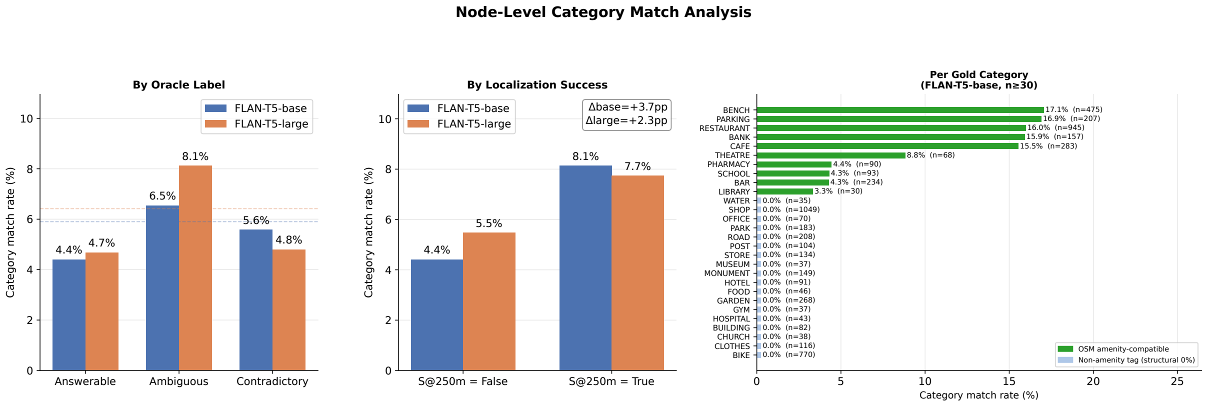

Figure 7: Node-level category match analysis. **Left:** Similar match rates across oracle labels (4.4–8.1%) suggest model outputs are not conditioned on instruction structure. **Center:** Higher match rates for geographically correct predictions confirm positive correlation between geographic and category accuracy. **Right:** Per-category breakdown for FLAN-T5-base ($n \geq 30$). Green bars denote OSM `amenity`-compatible categories (e.g., BENCH: 17.1%); gray stubs denote categories under other OSM keys yielding zero matches by construction.

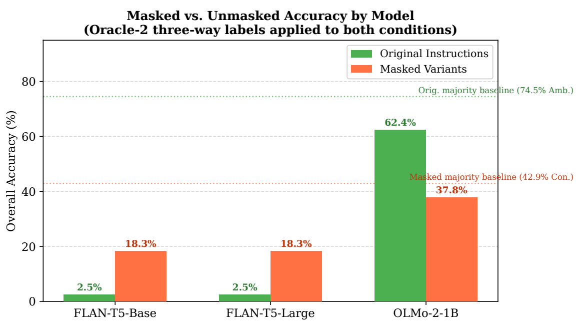

Figure 8: Overall accuracy for original vs. masked instructions. FLAN-T5 variants perform poorly across both conditions (max 18.3%), while OLMo-2-1B achieves higher accuracy (62.4% original, 37.8% masked), yet remains below the respective majority baselines of 74.5% and 42.9%.

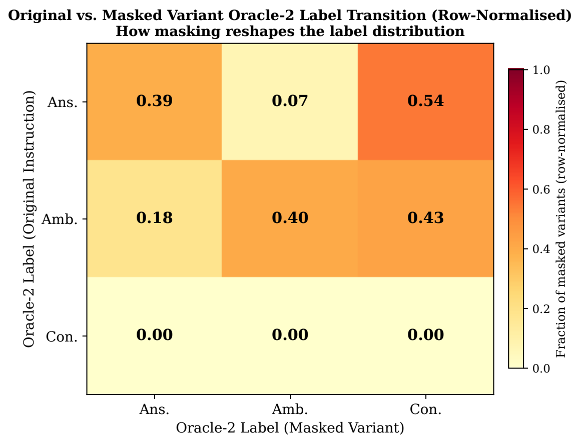

Figure 9: Row-normalised Oracle-2 label transition matrix (original → masked). *Contradictory* examples remain stable; most originally *Ambiguous* examples become *Contradictory* or remain *Ambiguous* after masking. Distinct from Table 2, which crosses Oracle-1 against Oracle-2 on originals.

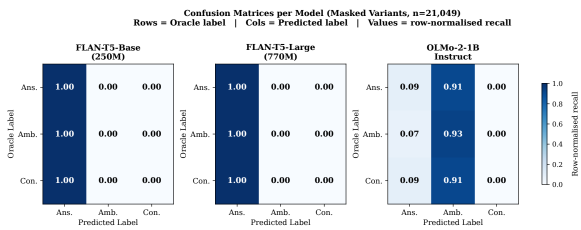

Figure 10: Confusion matrices for all three models on masked variants. FLAN-T5 predicts *Answerable* for nearly all inputs; OLMo predicts *Ambiguous* for the majority. *Contradictory* predictions remain rare across all models.

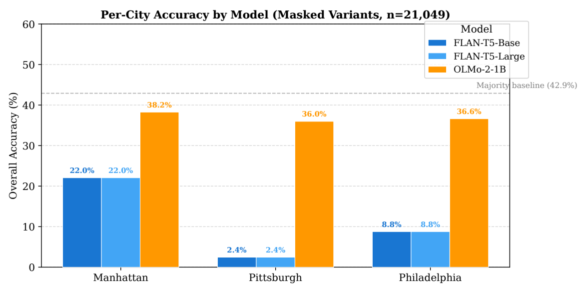

Figure 11: Per-city answerability accuracy for all models on masked variants. OLMo: Manhattan 38.2%, Philadelphia 36.6%, Pittsburgh 36.0%. FLAN-T5 shows near-total *Answerable* collapse in all cities. No city is an outlier.

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